

Graph Linear Canonical Transform: Definition, Vertex-Frequency Analysis and Filter Design

Jian Yi Chen , Yu Zhang , and Bing Zhao Li 

Abstract—This paper proposes a graph linear canonical transform (GLCT) by decomposing the linear canonical parameter matrix into fractional Fourier transform, scale transform, and chirp modulation for graph signal processing. The GLCT enables adjustable smoothing modes, enhancing alignment with graph signals. Leveraging traditional fractional domain time-frequency analysis, we investigate vertex-frequency analysis in the graph linear canonical domain, aiming to overcome limitations in capturing local information. Filter design methods, including optimal design and learning with stochastic gradient descent, are analyzed and applied to image classification tasks. The proposed GLCT and vertex-frequency analysis present innovative approaches to signal processing challenges, with potential applications in various fields.

Index Terms—Graph signal processing, graph linear canonical transform, vertex-frequency analysis, filter design.

I. INTRODUCTION

GRAPH signal processing has a broad spectrum of applications in computer vision, image processing, and social network analysis. Advancements in these technologies will catalyze technological innovation and societal progress, offering abundant developmental opportunities and benefits for humanity [1], [2], [3]. The essence of graph signal processing lies in the construction of dictionary atoms, followed by the representation of graph signals as linear combinations of these atoms [4]. This representation technique finds diverse applications, including data visualization analysis, statistical analysis, compression, and regularization in machine learning, and ill-posed inverse problems such as repair, denoising, and classification tasks.

When the atom is the eigenvector (Fourier basis) obtained from the Laplacian matrix decomposition of the graph, different

Fourier bases correspond to distinct levels of smoothness. Their associated eigenvalues provide a gradient characterization of the smoothness of the graph signal, effectively functioning as frequencies [5], [6]. Lower eigenvalues indicate slower changes in the corresponding Fourier basis, resulting in consistent signal values across similar nodes. Conversely, higher eigenvalues signify more rapid changes in the Fourier basis, leading to inconsistent signal values across similar nodes. Thus, the coefficients of the graph Fourier transform (GFT) represent the amplitude of the graph signal on the corresponding frequency component, reflecting the strength of the signal on that frequency component.

Based on the GFT, various graph analysis methods have emerged, with the core focus being on designing different dictionary atoms to represent graph signals. The effectiveness of signal analysis refers to the ability to capture important information in a signal with fewer mathematical descriptions [7], which essentially requires atoms to be able to sparsely represent the signal. Therefore, the effectiveness of vertex-frequency analysis of graph signals requires atoms to have the following characteristics in the spatial domain:

- **Interpretability:** The methods should enable the explanation of the underlying graph structure.
- **Multiresolution:** They should be able to continuously approximate signals from low-frequency resolution to high-frequency resolution.
- **Locality:** Emphasis should be placed on the local structure of the graph.
- **Multiscale:** They should encompass vertex-frequency distributions of various scales.
- **Smooth diversity:** The ability to select different smoothing modes should be included.

Traditional vertex-frequency analysis methods can effectively fulfill the first four characteristics but are inadequate in addressing the last one [8], [9], [10]. The fixed smoothing mode based on the GFT relies solely on its Laplacian matrix. As a result, traditional vertex-frequency analysis methods are limited to characterizing signals from the horizontal and vertical directions of the vertex-frequency plane [11], [12], [13]. However, due to the diverse and complex nature of graph signals, the GFT-based processing method encounters limitations; specifically, a single smoothing mode is insufficient to represent these signals effectively. Traditional vertex-frequency analysis approaches have not fully leveraged the inherent geometric characteristics of graph signals, resulting in performance bottlenecks and

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challenges in achieving locally optimal approximations in graph signal processing. To address these limitations, a new vertex-frequency analysis method is needed.

The linear canonical transform (LCT) is a generalized integral transform encompassing the Fourier transform (FT), fractional Fourier transform (FrFT), and Fresnel transform as special cases [14], [15]. Its additional parameters offer strong flexibility. Building upon the advantages of the LCT, it has been further integrated into graph signal processing technology, leading to significant research achievements. These include the development of graph fractional Fourier transform (GFrFT) [16], graph linear canonical transform (GLCT) [17] based on adjacency matrix, and related vertex-frequency analysis methods [18], [19], [20], [21]. The linear canonical parameter matrix can be decomposed into a linear combination of FrFT, scale transform, and chirp modulation (CM). Leveraging this pivotal property, this paper proposes a GLCT based on the graph Laplacian matrix, which significantly diverges from previous work [17]. The GLCT facilitates the adjustment of the smoothing mode to better align with the graph signal. Additionally, building upon the research foundation of traditional fractional domain time-frequency analysis methods [22], [23], [24], [25], this paper explores vertex-frequency analysis methods in the graph linear canonical domain. This research endeavors to address the inherent limitations of GLCT techniques in characterizing local information.

The vertex-frequency analysis method based on GLCT allows atoms to select different smoothing modes, enabling optimal matching with the signal's intrinsic local features and offering new approaches in signal processing. This paper introduces a systematic framework for GLCT-based vertex-frequency analysis and discusses three filter definitions: (1) A windowed GLCT, defined through the convolution and translation operators of GLCT; (2) The graph linear canonical wavelet and Stockwell transforms, defined by the spectral shift of the bandpass function kernel in the graph's canonical domain; (3) A local GLCT, defined by constructing a filter based on the vertex neighborhood. Selecting the appropriate filter is essential in practical applications, especially as different spectral neural networks rely on various filter learning methods [26], [27], [28], [29]. This paper examines filter design approaches, including optimal filter design and stochastic gradient descent-based filter learning, and applies these filters to image classification tasks. The proposed filters can serve as convolutional layers in spectral graph neural networks [30], [31], [32], [33], [34], demonstrating strong potential for diverse applications.

This paper is structured through the following steps. Section II presents the fundamental theory of graph signal processing and the LCT. In Section III, the GLCT is introduced through the use of the GFrFT, graph scaling, and the graph CM method, with a discussion on the significance of the definition and the impact of various parameters. In Section IV, the definition of the vertex-frequency analysis method based on GLCT is discussed and summarized. In Section V, we focus on optimal filter design and filter learning, applying the proposed filters to image classification tasks. Finally, Section VI concludes this paper.

II. PRELIMINARIES

A. The Graph Signal

Consider a weighted, undirected, and connected graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{W}\}$, where $\mathcal{V} = \{v_0, v_1, \dots, v_{N-1}\}$ with $|\mathcal{V}| = N$ represents a finite set of vertices, $\mathcal{E}$ denotes the set of edges, and $\mathbf{W}$ represents the weighted adjacency matrix. In this graph, if there exists an edge $e = (i, j)$ connecting vertices i and j , the weight of the edge is denoted by $w_{i,j}$; otherwise, $w_{i,j} = 0$ [5].

The non-normalized graph Laplacian is a symmetric difference operator $\mathcal{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ represents a diagonal degree matrix of the graph $\mathcal{G}$. The i -th diagonal element d_i of $\mathbf{D}$ is equal to the sum of the weights of all the edges incident to vertex i . Since the matrix $\mathcal{L}$ is real and symmetric, it possesses a complete set of orthonormal eigenvectors denoted as $\{\mathbf{u}_k\}_{k=0,1,\dots,N-1}$. These eigenvectors are sorted in ascending order according to their corresponding eigenvalues $0 = \tilde{\lambda}_0 < \tilde{\lambda}_1 \leq \tilde{\lambda}_2 \leq \dots \leq \tilde{\lambda}_{N-1}$. Thus, using the unitary column matrix $\mathbf{U} = [\mathbf{u}_0, \mathbf{u}_1, \dots, \mathbf{u}_{N-1}]$ and the diagonal matrix $\mathbf{\Lambda} = \text{diag}([\tilde{\lambda}_0, \tilde{\lambda}_1, \dots, \tilde{\lambda}_{N-1}])$, we can express

$$\mathcal{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H \quad (1)$$

where the superscript H represents the Hermitian transpose operation. While we use the non-normalized graph Laplacian $\mathcal{L}$ for explanation in this article, we can still use the normalized graph Laplacian $\mathcal{L}_{\text{norm}} = \mathbf{I}_N - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, where $\mathbf{I}_N$ represents an N -dimensional identity matrix [4].

A signal defined on the graph $\mathcal{G}$ can be represented as a mapping [19]

$$\begin{aligned} f: \mathcal{V} &\rightarrow \mathbb{R} \\ v_i &\mapsto f(i). \end{aligned} \quad (2)$$

Alternatively, we can express $\mathbf{f}$ as a real-valued vector $\mathbf{f} = [f(0), f(1), \dots, f(N-1)]^T \in \mathbb{R}^N$, where $f(i)$ represents the value associated with the i -th vertex. The Laplacian matrix is an operator that captures the local smoothness of a graph signal. When it acts on the graph signal f , it is defined as

$$(\mathcal{L}\mathbf{f})(i) = \sum_{j \in \mathcal{N}_i} w_{i,j} [f(i) - f(j)] \quad (3)$$

where the neighborhood $\mathcal{N}_i$ represents the set of vertices connected to vertex i by an edge. This operator quantifies the difference in signal between the central node and its neighboring nodes.

Example 1: We simulate two distinct graphs: a 128-node sensor network graph and a 256-node Swiss roll graph. We consider the random signal for both cases, as shown in Fig. 1(a) and 1(b), respectively.

B. The Graph Fractional Fourier Transform

1) *Graph Fourier Transform:* For any function $\mathbf{f} \in \mathbb{R}^N$ defined on the vertices of $\mathcal{G}$, its GFT is defined as [5]

$$\hat{f}(k) = \langle f, \mathbf{u}_k \rangle = \sum_{n=0}^{N-1} f(n) \mathbf{u}_k^*(n), \quad k = 0, 1, \dots, N-1. \quad (4)$$

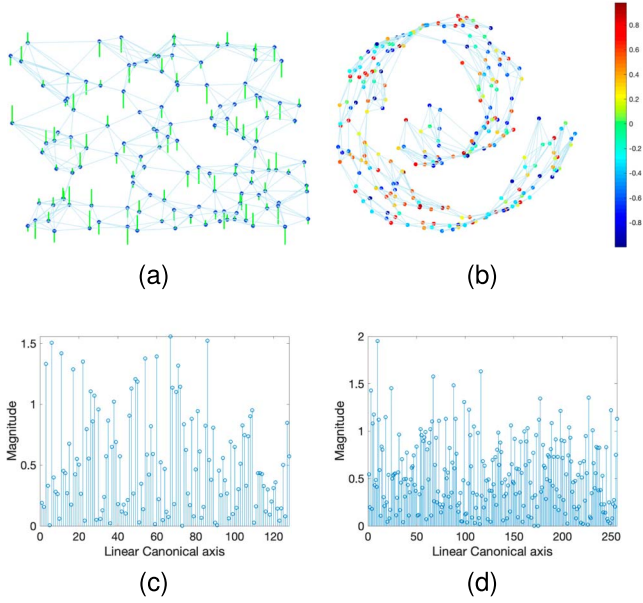


Fig. 1. (a) A random signal on a sensor network graph. (b) A random signal on the Swiss roll graph. (c) The GLCT with $\sigma = 1/3$, $\xi = 0$, $\beta = 0.8$ of (a). (d) The GLCT with $\sigma = 1/3$, $\xi = 0$, $\beta = 0.8$ of (b).

The inverse GFT is given by

$$f(n) = \sum_{k=0}^{N-1} \hat{f}(k) u_k(n), \quad n = 0, 1, \dots, N-1. \quad (5)$$

Therefore, we can observe that the GFT can be represented using the matrix $\mathbf{U}$, such that $\hat{\mathbf{f}} = \mathcal{F}\mathbf{f} = \mathbf{U}^H \mathbf{f}$ [3]. In the case where the GFT matrix $\mathcal{F}$ is ortho-diagonalized, it can be expressed as $\mathcal{F} = \mathbf{U}^H = \mathbf{Q}\mathbf{R}\mathbf{Q}^H$, where the matrices $\mathbf{Q} = [\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_{N-1}]$ and $\mathbf{R} = \text{diag}([r_0, r_1, \dots, r_{N-1}])$ are obtained through the spectral decomposition of the GFT matrix.

2) *Graph Fractional Fourier Transform*: By denoting $\mathbf{J} = \text{diag}([\zeta_0, \zeta_1, \dots, \zeta_{N-1}]) = \mathbf{R}^\alpha$, where $\zeta_l = r_l^\alpha$, the GFrFT is defined as $\hat{\mathbf{f}}_\alpha = \mathcal{F}_\alpha \mathbf{f} = \mathbf{Q}\mathbf{J}\mathbf{Q}^H \mathbf{f}$ [16]. There is [19]

$$\hat{f}_\alpha(k) = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} f(n) q_l(k) \zeta_l q_l^*(n) \quad (6)$$

where the parameter α is the order of the FrFT, and $k = 0, 1, \dots, N-1$. Similar to (4), we define the graph fractional Laplacian operator $\mathcal{L}_\alpha$ as follows

$$\mathcal{L}_\alpha = \mathcal{F}_\alpha^H \mathbf{L} \mathcal{F}_\alpha. \quad (7)$$

The inverse GFrFT is given by

$$f(n) = \sum_{k=0}^{N-1} \hat{f}_\alpha(k) q_l^*(k) \zeta_l q_l(n), \quad n = 0, 1, \dots, N-1. \quad (8)$$

C. Linear Canonical Transform

The linear canonical transform (LCT) of the signal $f(t) \in L^2(\mathbb{R})$ with the parameters $A = (a, b, c, d)$, where $ad - bc = 1$,

is defined as follows [14], [15], [35]

$$L^A\{f\}(u) = \begin{cases} \int_{-\infty}^{\infty} f(t) K_A(t, u) dt & b \neq 0 \\ \sqrt{d} e^{i(c d/2) u^2} f(dt) & b = 0 \end{cases} \quad (9)$$

$$K_A(t, u) = \sqrt{\frac{1}{i 2 \pi b}} e^{i(\frac{d}{b} u^2 - \frac{2}{b} u t + \frac{a}{b} t^2)} \quad (10)$$

where i represents an imaginary unit. When the parameter $A = (\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha)$ is used, the LCT reduces to the α^{th} order FrFT, as follows:

$$L^{(\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha)}\{f\}(u) = \int_{-\infty}^{\infty} K_\alpha(t, u) f(t) dt$$

$$K_\alpha(t, u) = \begin{cases} A_\alpha \exp \left\{ i \frac{u^2 \cot \alpha}{2} - i t u \csc \alpha + i \frac{t^2 \cot \alpha}{2} \right\} & \alpha \neq n\pi \\ \delta(u - t), \alpha = 2n\pi \\ \delta(u + t), \alpha = 2n\pi - \pi \end{cases}$$

$$A_\alpha = \sqrt{\frac{1 - i \cot(\alpha)}{2\pi}}, \quad 0 < |\alpha| < \pi. \quad (11)$$

The LCT with parameter $A = (a, b, c, d)$ can be decomposed into a combination of various specialized transforms, achieved through the decomposition of the parameter matrix. This article will utilize the following decomposition

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \xi & 1 \end{bmatrix} \begin{bmatrix} \sigma & 0 \\ 0 & \sigma^{-1} \end{bmatrix} \begin{bmatrix} \cos \beta \pi / 2 & \sin \beta \pi / 2 \\ -\sin \beta \pi / 2 & \cos \beta \pi / 2 \end{bmatrix} \quad (12)$$

where A is decomposed into FrFT, scaling, and chirp modulation (CM), ξ is the CM parameter, σ denotes the scaling parameter, and β is the FrFT parameter. The parameter relations between (a, b, c, d) and (ξ, σ, β) are $\xi = \frac{ac+bd}{a^2+b^2}$, $\sigma = \sqrt{a^2+b^2}$, $\beta = \frac{2}{\pi} \cos^{-1} \left(\frac{a}{\sigma} \right) = \frac{2}{\pi} \sin^{-1} \left(\frac{b}{\sigma} \right)$. In Section III, we will present the definition of graph linear canonical transform (GLCT) based on this parameter matrix factorization.

III. GRAPH LINEAR CANONICAL TRANSFORM

A. Definition

Building upon the centrality and scalability of the LCT eigendecomposition approach, this article proposes a GLCT framework based on the Laplacian matrix by decomposing the LCT parameter matrix in (12). Since any LCT matrix is non-commutative, the implementation should follow the order of FrFT, scaling, and CM. The definition of the GLCT based on the Laplacian matrix is derived through the following three stages.

Step1: GFrFT stage

For a graph signal $\mathbf{f}$, initiate the GFrFT as delineated in Section II-B2, expressed as

$$\hat{\mathbf{f}}_\beta = \mathcal{F}_\beta \mathbf{f} = \mathbf{Q} \mathcal{K} \mathbf{Q}^H \mathbf{f} \quad (13)$$

where $\mathcal{K} = \text{diag}([\kappa_0, \kappa_1, \dots, \kappa_{N-1}]) = \mathbf{R}^\beta$, $\kappa_l = r_l^\beta$, $l = 0, \dots, N-1$, and $\beta = \frac{2}{\pi} \cos^{-1} \left(\frac{a}{\sigma} \right) = \frac{2}{\pi} \sin^{-1} \left(\frac{b}{\sigma} \right)$.

Step2: Graph scaling stage

Similar to the definition of the GFrFT, we begin with Hermite-Gaussian functions to define the graph scale transform. In classical signal processing, a scaled signal can be obtained by first decomposing the signal into the undilated Hermite eigenspace and then reconstructing it using dilated Hermite eigenvectors, while preserving the original inner product coefficients [36]. Specifically, this can be expressed as $f(n) = \sum_{k=0}^{N-1} c_k \Phi_k(n)$ and $f_\sigma(n) = \sum_{k=0}^{N-1} c_k \Phi_{k,\sigma}(n)$, where $\Phi_k(n)$ represents the undilated discrete Hermite functions of order k , and $\Phi_{k,\sigma}(n)$ denotes the discrete Hermite functions of order k with the scaling parameter σ .

For the graph $\mathcal{G}$, if the graph Fourier operator is defined as $\mathcal{F} = \mathbf{U}^H = \mathbf{Q}\mathbf{R}\mathbf{Q}^H$, then we define a scaling matrix as

$$\mathbf{S}_\sigma = \text{diag}([\sigma^{-\varepsilon r_0}, \sigma^{-\varepsilon r_1}, \dots, \sigma^{-\varepsilon r_{N-1}}]) \quad (14)$$

where r_l , $l = 0, \dots, N-1$, represents the eigenvalues corresponding to the undilated eigenspace $\mathbf{Q}$, and ε is a factor that represents the rate of scale change. A detailed explanation of the scaling matrix can be found in Appendix A. The dilated graph eigenspace is defined as

$$\mathbf{P} = \mathbf{S}_\sigma \mathbf{Q} \quad (15)$$

where $\mathbf{P} = [\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_{N-1}]$. Similarly, a scaled graph signal $ST_\sigma \mathbf{f}$ can be obtained by first decomposing the signal into the undilated graph eigenspace and then reconstructing it using the dilated graph eigenspace. Therefore, the graph scale transform is defined as

$$ST_\sigma \mathbf{f} = \mathbf{P}\mathbf{Q}^H \mathbf{f} \quad (16)$$

where $\sigma = \sqrt{a^2 + b^2}$.

Step3: Graph CM stage

In classical signal processing, the modulation operator is defined by $M_\xi f = e^{i\pi \xi t} f(t)$. Shuman defined the modulation operator on the graph as $M_k f = f(n) u_k^2(n)$, where $k \in \{0, \dots, N-1\}$, recognizing that the essence of the modulation operator lies in multiplying by the Laplacian eigenvectors.

However, the scenario addressed in this article pertains to CM, where $CM_\xi f = e^{i\pi \xi t^2} f(t)$. The concept of CM on the graph has not yet been clearly defined, and defining it from a practical perspective is quite challenging. Hence, we define it based on the chirp signal spectrum perspective.

In traditional signal processing, the Fourier spectrum of a chirp signal $s(t) = \text{rect}(\frac{t}{T}) e^{i\pi \xi t^2}$ can be approximated as $F\{s\}(u) \approx \text{rect}(\frac{u}{|\xi|T}) e^{-i\pi \frac{u^2}{\xi}}$ [37]. Similarly, we define a signal in a graph spectral domain as

$$\hat{s}_\xi(k) = \begin{cases} e^{-i\frac{\lambda_k^2}{\xi}}, & \xi \neq 0 \\ \mathbf{1}_N, & \xi = 0 \end{cases} \quad (17)$$

where $\mathbf{1}_N$ represents an N -dimensional column vector with all elements being 1. Next, the inverse GFT is performed to obtain its vertex domain form as $s(n) = \mathbf{U} \hat{s}_\xi(k)$. We utilize this signal as the CM term for the graph signal. Let

$$\Upsilon = \text{diag}(\mathbf{U} \hat{s}_\xi(k)) \quad (18)$$

thus, defining the CM operator on the graph as

$$CM_\xi \mathbf{f} = \Upsilon \mathbf{f} \quad (19)$$

where $\xi = \frac{ac+bd}{a^2+b^2}$.

Based on the aforementioned three stages, we define the GLCT by performing FrFT in (13), scaling in (16), and CM in (19) on a graph signal $\mathbf{f}$.

Definition 1: For the parameter $A = (a, b, c, d)$ of LCT, denote $\sigma = \sqrt{a^2 + b^2}$, $\xi = \frac{ac+bd}{\sigma^2}$, $\beta = \frac{2}{\pi} \cos^{-1}(\frac{a}{\sigma}) = \frac{2}{\pi} \sin^{-1}(\frac{b}{\sigma})$. The graph linear canonical transform (GLCT) is defined by

$$\hat{\mathbf{f}}_A = \mathcal{F}_A \mathbf{f} = \Upsilon(\mathbf{P}\mathbf{Q}^H)(\mathbf{Q}\mathcal{K}\mathbf{Q}^H)\mathbf{f} = \Upsilon\mathbf{P}\mathcal{K}\mathbf{Q}^H \mathbf{f}. \quad (20)$$

Its algebraic form is

$$\hat{f}_A(k) = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} f(n) \epsilon_k p_l(k) \kappa_l q_l^*(n). \quad (21)$$

The corresponding eigenvalues are $\Lambda_A = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{N-1}])$, where $\lambda_l = \tilde{\lambda}_l^\beta / \sigma$, $l = 0, \dots, N-1$.

Likewise, when $A = (\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha)$ is used, the equation $\hat{f}_{A=(\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha)}(k) = \hat{f}_\alpha(k)$ holds true. Additionally, when $A = (0, 1, -1, 0)$, we have $\hat{f}_{A=(0,1,-1,0)}(k) = \hat{f}(k)$ since $\Upsilon\mathbf{P}\mathcal{K}\mathbf{Q}^H$ reduces to $\mathbf{Q}\mathbf{R}\mathbf{Q}^H$. We define graph linear canonical Laplacian operator as

$$\mathcal{L}_A = \mathcal{F}_A^H \Lambda_A \mathcal{F}_A. \quad (22)$$

The inverse GLCT (IGLCT) is defined as $\mathbf{f} = \mathcal{F}_A^H \hat{\mathbf{f}}_A = \mathbf{Q}\mathcal{K}^H \mathbf{P}^H \Upsilon^H \hat{\mathbf{f}}_A$, which is equal to

$$f(n) = \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} \hat{f}_A(k) q_l(n) \kappa_l^* p_l^*(k) \epsilon_k^*. \quad (23)$$

In Example 2, we present the results of applying the GLCT to random graph signals on a sensor network graph and a Swiss roll graph, using the parameters $\sigma = 1/3$, $\xi = 0$, and $\beta = 0.8$. In Section III-B, we will analyze and explain the principles of GLCT and the influence of its parameters.

Remark 1: Without loss of generality, this article utilizes the Laplacian operator for explanations, which can alternatively be replaced by the adjacency matrix $\mathbf{W}$, the row-normalized matrix $\mathbf{Q} = \mathbf{D}^{-1}\mathbf{W}$, or the normalized Laplacian $\mathcal{L}_{\text{norm}} = \mathbf{I}_N - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$.

Example 2: Perform the GLCT with parameters $\sigma = 1/3$, $\xi = 0$, and $\beta = 0.8$ on the graph and graph signal in Example 1. The results are depicted in Fig. 1(e) and 1(f), respectively.

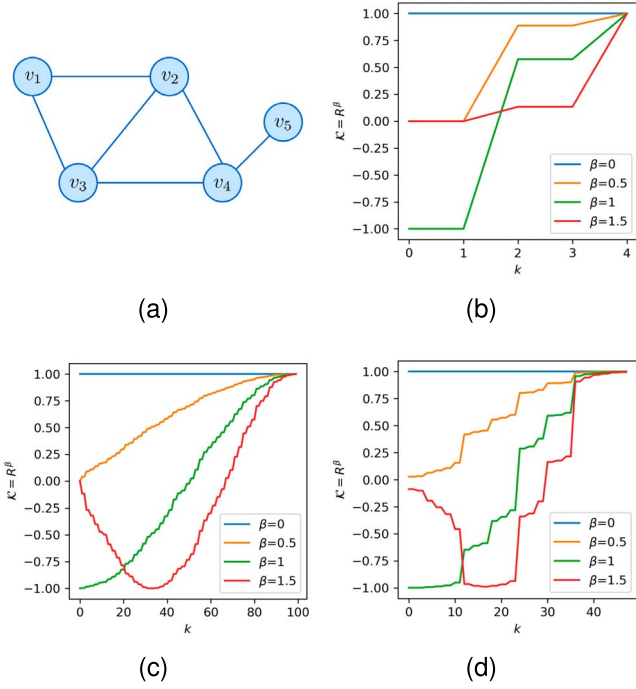


Fig. 2. (a) An undirected graph. (b) The effect of varying β on the eigenvalues of (a), with $K = R^\beta$. (c) The effect of varying β on the eigenvalues of the K-NN graph. (d) The effect of varying β on the eigenvalues of the path graph.

B. Explanation

In this section, we discuss the principles and effects of the three stages defined for GLCT. Firstly, numerous scholars have made significant research contributions to the GFrFT stage [16], [18], [19], [29]. The impact of the parameter β on the graph fractional Fourier operator is further investigated. Consider the undirected graph shown in Fig. 2(a). Fig. 2(b) illustrates the effect of varying β on the eigenvalues, with $K = R^\beta$. Additionally, a K-nearest neighbors (K-NN) graph with 100 vertices, where each vertex is connected to 4 neighbors, and a path graph with 48 vertices are analyzed. Fig. 2(c) and 2(d) demonstrate the impact of different β values on the eigenvalues, with $K = R^\beta$. All displayed values represent the real parts of the eigenvalues.

- $\beta > 1$: When $-1 < \Re(r_k) < 0$, $\Re(r_k)$ with larger absolute values will shrink, while smaller $\Re(r_k)$ will expand, where $\Re(\cdot)$ represents the real part. As $\Re(r_k)$ approaches 0, the phase oscillation becomes very pronounced. $\mathcal{F}_\beta$ will be more inclined towards low-frequency components.
- $\beta = 1$: $\mathcal{F}_\beta$ is equivalent to the graph Fourier operator $\mathcal{F}$.
- $0 < \beta < 1$: $\Re(r_k^\beta)$ is compressed overall, eliminating negative values. Consequently, the high-frequency components of $\mathcal{F}_\beta$ become more pronounced.
- $\beta = 0$: $\mathcal{F}_\beta$ becomes the identity matrix.

The specific value of β will significantly influence the properties of $\mathcal{F}_\beta$, resulting in varied effects across different application scenarios.

Secondly, the principles of the graph scale transform and the influence of its parameters are analyzed. In Appendix A, Fig. 19 is used as a case study to analyze the impact of varying values of σ on both the scale matrix and the dilated eigenspace.

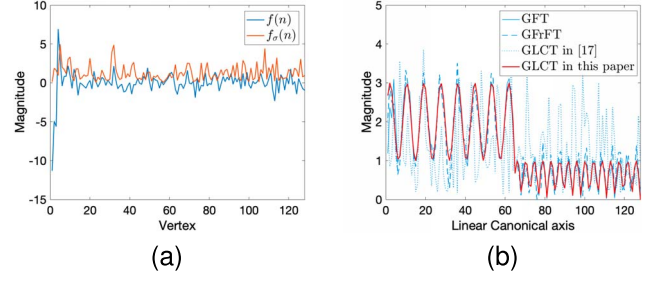


Fig. 3. (a) The graph signal $f(n)$ and the resulting dilated signal $f_\sigma(n)$. (b) The GFT, GFrFT ($\alpha = 0.9$), the GLCT from [17] ($\sigma = 1.73, \xi = 0, \beta = 1$), and the GLCT proposed in this paper ($\sigma = 1.73, \xi = 0, \beta = 1$) applied to $f_\sigma(n)$.

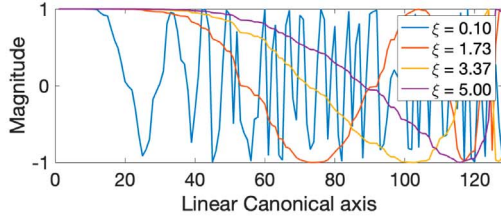
The influence of parameters in graph scale transform can be observed as follows:

- When $0 < \sigma < 1$, the scaled graph signal is compressed, resulting in a narrower waveform and faster oscillation.
- When $\sigma = 1$, the graph signal is not scaled.
- When $\sigma > 1$, the scaled graph signal expands, resulting in a wider waveform and slower oscillation.

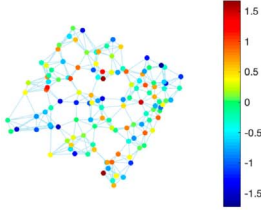
We have introduced a parameter ε to the scaling matrix, allowing adjustment of the degree of fluctuation. Typically, it is set to the default value of $\varepsilon = 0.5$. We also investigate the effects of the GLCT with varying values of σ on the graph signal. Consider the graph signal in the spectral domain as $\hat{f} = [21_{N/2}, 0_{N/2}]^\top + [\sin(\frac{30 \cdot n \cdot \pi}{N})]_{n=1}^N$, which is defined on the sensor network graph in Fig. 1(a). The corresponding graph signal in the vertex domain is $f(n)$, and its dilated signal is denoted as $f_\sigma(n)$, dilated by the eigenspace with $\sigma = 1.3$. Fig. 3(a) shows both $f(n)$ and the resulting $f_\sigma(n)$ under the influence of the graph-scale characteristic space with $\sigma = 1.3$. Next, various transformations are applied to $f_\sigma(n)$, as shown in Fig. 3(b). The figure displays $f_\sigma(n)$ under the GFT, GFrFT, the GLCT from [17], and the GLCT proposed in this paper. The results reveal that when the signal exhibits significant graph-scale characteristics (see Appendix A), the method introduced in this paper more effectively captures the frequency distribution characteristics of the graph signal.

Thirdly, the graph CM stage is examined. In traditional signal processing, CM is primarily used to increase bandwidth and resolution, as well as to enhance the accuracy of target detection and localization. A key application of CM is the dechirp process, which facilitates parameter estimation. The graph CM defined in this paper is expected to offer similar functionalities for graph signals. First, based on the definition, we analyze the impact of varying values of ξ on the graph spectral domain $\hat{s}_\xi(k)$ using the sensor network graph shown in Fig. 1(a), as illustrated in Fig. 4(a). The following observations can be made:

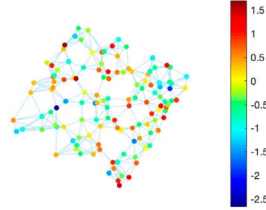
- When $\xi \neq 0$ is large: Larger values of ξ produce a slower and smoother variation in $\hat{s}_\xi(k)$, resulting in a graph spectrum dominated by low-frequency components and a more stable signal.
- When $\xi \neq 0$ is small: Smaller values of ξ cause the exponential term in $\hat{s}_\xi(k)$ to change more rapidly, resulting in more oscillations and the inclusion of higher-frequency components in the graph spectrum.



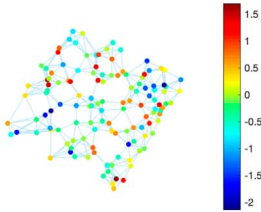
(a)



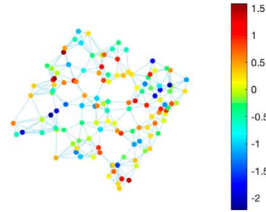
(b)



(c)



(d)

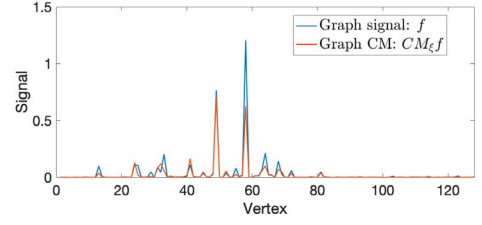


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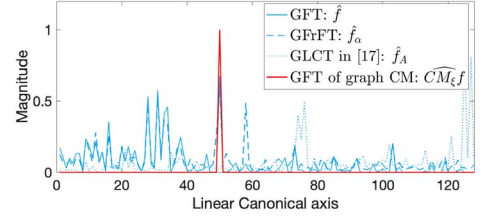
Fig. 4. (a) The impact of varying values of ξ on the graph spectral domain $\hat{s}_\xi(k)$ using the sensor network graph shown in Fig. 1(a). (b) $CM_\xi = Y, \xi = 0.1$. (c) $CM_\xi = Y, \xi = 1.73$. (d) $CM_\xi = Y, \xi = 3.37$. (e) $CM_\xi = Y, \xi = 5$.

Fig. 4(b)–4(e) show the corresponding $CM_\xi = Y$ for different values of ξ . Modulating ξ intuitively alters the smoothness of the plot, primarily affecting the high- and low-frequency components in the spectral domain. Consider a graph signal defined as $\mathbf{f} = \mathbf{U}^{-1} e^{i(1.73\lambda^2)} \mathbf{u}_{50}$, where $\mathbf{u}_{50}$ is the eigenvector in $\mathbf{U}$. Let $\xi = 1.73$, and apply the graph CM to this signal. The results are shown in Fig. 5(a). In Fig. 5(b), both the original and chirp-modulated signals are presented in the graph spectral domain, demonstrating that graph CM can achieve an effect similar to traditional signal processing dechirping.

Finally, the influence of GLCT parameters on the transform matrix is illustrated using Fig. 2(a) as an example. Fig. 6 shows the eigenvectors under varying GLCT parameters. It can be observed that adjusting these parameters alters the smoothing patterns of the eigenvectors, thereby allowing for improved fitting of different graph signals. For example, with parameters $\sigma = 0.5$, $\xi = 2.5$, and $\beta = 0.8$, the eigenvectors of the GLCT exhibit relatively low overall smoothness. This set of bases is suitable for fitting graph signals that exhibit rapid waveform oscillations. Example 3 presents specific cases demonstrating how to achieve optimal representation of graph signals by adjusting the parameters of the GLCT. This example highlights

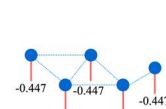


(a)

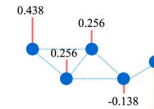


(b)

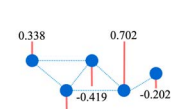
Fig. 5. (a) The graph signal $\mathbf{f} = \mathbf{U}^{-1} e^{i(1.73\lambda^2)} \mathbf{u}_{50}$ and the graph CM $CM_\xi \mathbf{f}$ with $\xi = 1.73$. (b) The graph spectrum of $\mathbf{f}$ and $CM_\xi \mathbf{f}$.



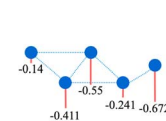
(a)



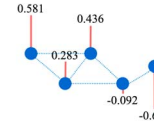
(b)



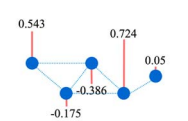
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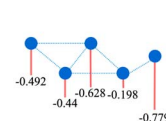
(d)



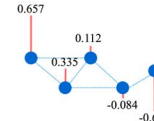
(e)



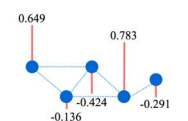
(f)



(g)



(h)



(i)

Fig. 6. Use vectors as graph signals. (a)–(c) The 0th, 1th, and 4th eigenvectors of the GLCT with $\sigma = 1$, $\xi = 0$, $\beta = 1$ (GFT). (d)–(f) The 0th, 1th, and 4th eigenvectors of the GLCT with $\sigma = 1$, $\xi = 0$, $\beta = 0.8$ (GFrFT). (g)–(i) The 0th, 1th, and 4th eigenvectors of the GLCT with $\sigma = 0.5$, $\xi = 2.5$, $\beta = 0.8$.

the advantages of using different smoothing modes, as these modes can adapt to various signals.

Example 3: Consider the graph signal $f(n) = 0.2\Re(\varsigma_{10}(n)) + 3\Re(\varsigma_{40}(n))$ for the sensor network graph in Example 1, where $\varsigma_k(n) = \sum_{l=0}^{N-1} \epsilon_k p_l(k) \kappa_l q_l^*(n)$ is derived from the graph linear canonical Laplacian matrix of the sensor network graph with $\sigma = 0.5$, $\xi = 2.5$, and $\beta = 0.8$. Fig. 7 shows the results of the GFT, GFrFT, and GLCT with different parameters for this signal. Adjusting the GLCT parameters allows adaptation to more diverse and complex graph signals.

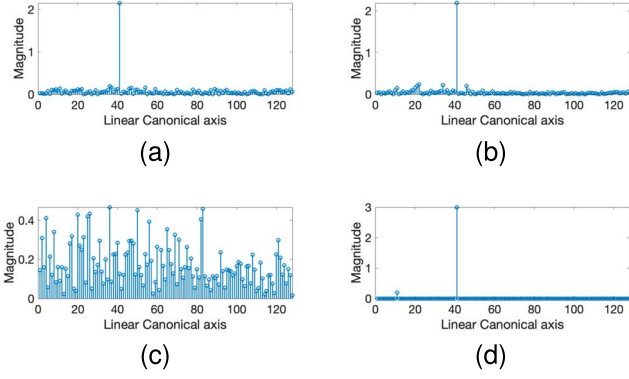


Fig. 7. Different transforms of $f(n) = 0.2\Re(\varsigma_{10}(n)) + 3\Re(\varsigma_{40}(n))$. (a) GFT. (b) GFrFT with $\alpha = 0.9$. (c) GLCT in [17] with $\sigma = 0.5$, $\xi = 2.5$, $\beta = 0.8$. (d) GLCT in this paper with $\sigma = 0.5$, $\xi = 2.5$, $\beta = 0.8$.

IV. VERTEX-FREQUENCY ANALYSIS METHOD ASSOCIATED WITH GLCT

Like classical time-frequency analysis, the spectral transform of signals localized around a specific vertex i serves as the fundamental framework for vertex-frequency analysis. This spectral transform is typically achieved using a localization window. Hence, a generalized method for vertex-frequency analysis, denoted as VGLCT, can be derived by combining the GLCT with local window functions [8], [9]. The VGLCT is obtained by computing the GLCT of a signal $f(n)$ and multiplying it by a suitable vertex localization window function, $h_i(n)$, resulting in the following expression

$$\mathcal{V}_h^A f(i, k) = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} f(n) h_i(n) \epsilon_k p_l(k) \kappa_l q_l^*(n). \quad (24)$$

In general, the VGLCT initially localizes the signal around vertex i using the graph window function and then evaluates its GLCT performance. Specifically, the graph window function should have a value close to 1 at vertex i , indicating that the vertex is in close proximity to vertex i or its nearest neighbors. Conversely, for vertices that are distant from vertex i , the graph window function should have a value close to 0, indicating negligible influence on the analysis.

We can categorize the methods for constructing the graph window into three categories:

- The window function is constructed based on the convolution and translation operators of GLCT, leading to the definition of windowed GLCT.
- By utilizing the spectral shift form of the kernel of the bandpass function in the linear canonical domain of the graph, we can define graph linear canonical wavelet transform and graph linear canonical S-transform.
- A window function constructed based on the vertex neighborhood can define local GLCT.

A. Windowed Graph Linear Canonical Transform

Unlike the traditional FT, the windowed FT (or short-time FT) divides the signal into a series of short-time windows and

applies the FT to the signal within each window to obtain the spectral information of each window. In graph signal processing, the windowed GFT (WGFT) has also been proposed [10], [11]. The basic idea is to define the WGFT as the inner product of the signal f and a shifted and modulated window function h . The WGFT is defined by incorporating convolution, translation, and modulation operations into the graph signal [19]. In this section, we begin by defining the graph linear canonical convolution operator and the graph linear canonical translation operator. Subsequently, we define the windowed graph linear canonical convolution transform (WGLCT) by multiplying the signal f with the translated window function prior to applying the GLCT.

Graph linear canonical convolution operator. Consider two signals, $f(n)$ and $h(n)$, defined on a graph, with their corresponding GLCTs represented as $\hat{f}_A$ and $\hat{h}_A$. The graph linear canonical convolution $*_A$ of these signals is defined as follows

$$(f *_A h)(n) = IGLCT\{\hat{f}_A(k) \hat{h}_A(k)\} \\ = \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} \hat{f}_A(k) \hat{h}_A(k) q_l(n) \kappa_l^* p_l^*(k) \epsilon_k^*. \quad (25)$$

The concept of ‘shift’ on a graph cannot be directly and uniquely extended to graph signals using an analogy with classical signal shifts. In the realm of vertex frequency analysis, there has been an intriguing effort to generalize convolution and define corresponding shift operators on graphs. In this context, we aim to further generalize the calculation of shifts based on graph linear regularization transforms.

Graph linear canonical translation operator. For any signal $h(n)$ defined on a graph, and the delta function located at a vertex $i \in \{0, 1, \dots, N-1\}$, given by $\delta_i(n) = \delta(n-i)$. The GLCT of delta function $\delta_i(n)$, is given by

$$\Delta_i(k) = GLCT\{\delta_i(n)\} = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} \delta_i(n) \epsilon_k p_l(k) \kappa_l q_l^*(n) \\ = \sum_{l=0}^{N-1} \epsilon_k p_l(k) \kappa_l q_l^*(i). \quad (26)$$

The graph linear canonical translation T_i^A is defined as

$$T_i^A h(n) = (h *_A \delta_i)(n) = \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} \hat{h}_A(k) \\ \times \left(\sum_{s=0}^{N-1} \epsilon_k p_s(k) \kappa_s q_s^*(i) \right) q_l(n) \kappa_l^* p_l^*(k) \epsilon_k^*. \quad (27)$$

Therefore, the WGLCT can be written as

$$\mathcal{V}_{h,1}^A f(i, k) = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} f(n) T_i^A h^*(n) \epsilon_k p_l(k) \kappa_l q_l^*(n) \quad (28)$$

where $T_i^A h(n)$ is defined in (27). By applying the graph window function $T_i^A h(n)$ to the signal $f(n)$ on each vertex, and

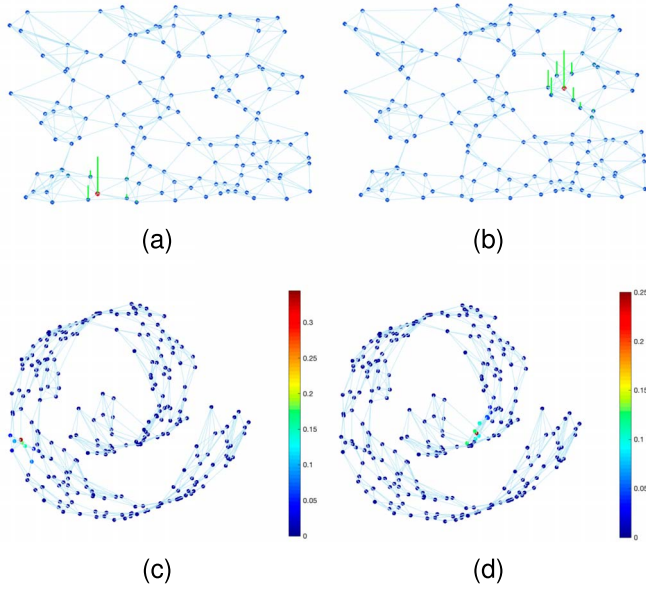


Fig. 8. GLCT with $\sigma = 1.1$, $\xi = 0$, and $\beta = 0.9$. The red dot in (a) (b) represents the selected vertex i , while the green line represents the magnitude of the function value. (a) $T_{30}^A h$ on sensor network graph. (b) $T_{90}^A h$ on sensor network graph. (c) $T_{30}^A h$ on swiss roll graph. (d) $T_{200}^A h$ on swiss roll graph.

then summing over all vertices i , a constant overlap-add relation is achieved. Then the inverse WGLCT is

$$f(n) = \frac{1}{\sum_{i=0}^{N-1} T_i^A h(n)} \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} \mathcal{V}_{T_i^A h}^A f(i, k) q_l(n) \kappa_l^* p_l^*(k) \epsilon_k^*. \quad (29)$$

The basic form of this window, $h(n)$, can be conveniently defined in the spectral domain, for example as $H(k) = ce^{-\lambda_k \tau}$, where c denotes the window amplitude and $\tau > 0$ is a constant which determines the window width.

Example 4: Let $H(k) = e^{-\lambda_k/2}$ be the spectral domain-defined window, with $\sigma = 1.1$, $\xi = 0$, and $\beta = 0.9$. For the sensor network graph in Example 1, local vertex-frequency analysis is performed on vertex 30 (Fig. 8(a)) and vertex 90 (Fig. 8(b)), respectively. For the Swiss roll graph in Example 1, local vertex-frequency analysis is conducted on vertex 30 (Fig. 8(c)) and vertex 200 (Fig. 8(d)), respectively.

Example 5: Consider the graph signal as $f_1(n) = \Re(\varsigma_{30}(n)) + \Re(\varsigma_{80}(n))$ for the sensor network graph in Example 1, where $\varsigma_k(n) = \sum_{l=0}^{N-1} \epsilon_k p_l(k) \kappa_l q_l^*(n)$ is derived from the graph linear canonical Laplacian matrix of the sensor network graph. Let $H(k) = e^{-400\lambda_k}$, $\sigma = 1.1$, $\xi = 0$, and $\beta = 0.9$, the WGLCT of $f_1(n)$ is depicted in Fig. 9(a).

Example 6: Consider a signal on the Swiss roll graph in Example 1, $f_2(n) = \Re(\varsigma_{20}(n)) + \Re(\varsigma_{50}(n)) + \Re(\varsigma_{70}(n))$, where $\varsigma_k(n)$ is derived from the GLCT of the Swiss roll graph. Let $H(k) = e^{-3500\lambda_k}$, $\sigma = 1.1$, $\xi = 0$, and $\beta = 0.8$, the WGLCT of $f_2(n)$ is depicted in Fig. 9(b).

Next, we present a rapid computation approach for WGLCT. In the traditional scenario of 1D signals, the windowed FT (WFT) can be computed by taking the inverse FT from the

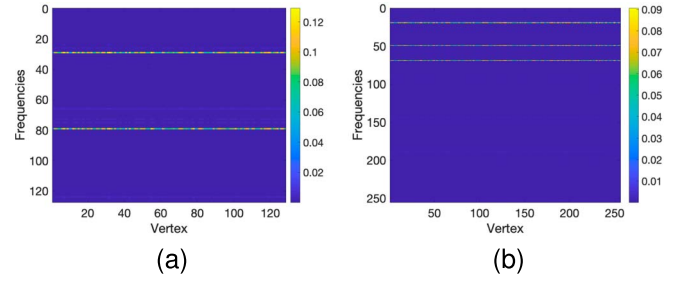


Fig. 9. The WGLCT of graph signals with $\sigma = 1.1$, $\xi = 0$, and $\beta = 0.9$. (a) The WGLCT of $f_1(n)$ on the sensor network graph. (b) The WGLCT of $f_2(n)$ on the Swiss roll graph.

so-called α -domain [19], [22], [38]. We define α -domain in the linear canonical domain by

$$\alpha(u, u') = \int_{-\infty}^{\infty} V_h^A f(l, u) K_A(l, u') dl \quad (30)$$

where $V_h^A f(l, u)$ represents the windowed LCT of f [25], that is, (30) applies LCT to $V_h^A f(l, u)$. The fast WLCT can be obtained by applying inverse LCT to α on the variable u'

$$FV_h^A f(l, u) = \int_{-\infty}^{\infty} \alpha(u, u') K_{A^{-1}}(l, u') du'. \quad (31)$$

Similarly, the α -domain is defined as the GLCT of WGLCT in the graph linear canonical domain as follows

$$\alpha(k, k') = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} \mathcal{V}_{T_i^A h}^A f(n, k') \epsilon_k p_l(k) \kappa_l q_l^*(n). \quad (32)$$

Then, the discretization of α is

$$\begin{aligned} \alpha(k, k') &= \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} \sum_{m=0}^{N-1} \sum_{r=0}^{N-1} f(m) \sum_{d=0}^{N-1} \sum_{y=0}^{N-1} \sum_{s=0}^{N-1} \hat{h}_A^*(d) \\ &\quad \times q_s(n) \kappa_s^* p_s^*(d) \epsilon_d^* \epsilon_d p_y(d) \kappa_y q_y^*(m) \\ &\quad \times \epsilon_{k'} p_r(k') \kappa_r q_r^*(m) \epsilon_k p_l(k) \kappa_l q_l^*(n) \\ &= \sum_{m=0}^{N-1} \sum_{r=0}^{N-1} \sum_{y=0}^{N-1} f(m) \hat{h}_A(k) \epsilon_{k'} p_r(k') \kappa_r q_r^*(m) \\ &\quad \times \epsilon_k p_y(k) \kappa_y q_y^*(m). \end{aligned} \quad (33)$$

The fast WGLCT is achieved by performing the inverse GLCT on the variable k with respect to α , that is

$$\begin{aligned} FV_{h,1}^A f(i, k') &= \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} \alpha(k, k') q_l(n) \kappa_l^* p_l^*(k) \epsilon_k^* \\ &= \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} \tilde{f}(k, k') \hat{h}_A(k) q_l(n) \kappa_l^* p_l^*(k) \epsilon_k^* \end{aligned} \quad (34)$$

where

$$\begin{aligned} \tilde{f}(k, k') &= \sum_{m=0}^{N-1} \sum_{r=0}^{N-1} \sum_{y=0}^{N-1} f(m) \epsilon_{k'} p_r(k') \kappa_r q_r^*(m) \\ &\quad \times \epsilon_k p_y(k) \kappa_y q_y^*(m). \end{aligned} \quad (35)$$

Algorithm 1 Fast windowed graph linear canonical transform**Input:**Graph signal f .An $N \times N$ matrix $\mathbf{F}$ (each row of $\mathbf{F}$ is the signal f).**Output:**The fast WGLCT matrix $\mathbf{FV}_{h,2}^A$.

- 1: Compute $\mathcal{F}_A = \mathbf{Y}\mathbf{P}\mathbf{K}\mathbf{Q}^H$.
- 2: Calculate $\tilde{f}(k, k')$ using the expression $\tilde{\mathbf{F}} = [\tilde{f}(k, k')] = (\mathbf{F} \circ \mathcal{F}_A^H) \cdot \mathcal{F}_A$, where $\circ$ represents element-wise multiplication (Hadamard multiplication), and $\cdot$ denotes standard matrix multiplication.
- 3: Construct a matrix where each row corresponds to the GLCT of the window function

$$\mathbf{H}_A = \begin{pmatrix} \hat{h}_A(r_0) & \hat{h}_A(k_1) & \dots & \hat{h}_A(k_{N-1}) \\ \hat{h}_A(k_0) & \hat{h}_A(k_1) & \dots & \hat{h}_A(k_{N-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{h}_A(k_0) & \hat{h}_A(k_1) & \dots & \hat{h}_A(k_{N-1}) \end{pmatrix}.$$
- 4: Compute the α -domain representation by multiplying $\tilde{\mathbf{F}}$ and $\mathbf{H}_A^*$, i.e., $\alpha = \mathbf{H}_A^* \cdot \tilde{\mathbf{F}}$.
- 5: The fast WGLCT at each vertex is obtained by applying inverse GLCT to σ , i.e., $\mathbf{FV}_{h,2}^A = \mathcal{F}_A \cdot \alpha$.

indicate that the efficiency of the fast algorithm has improved by approximately 79-fold.

B. Graph Linear Canonical Domain Localization

The classical time-frequency analysis method uses windows for frequency localization in the spectral domain, allowing the dual form to be derived using the DFT of the original signal and the spectral window. Similarly, for graph signals, this approach can achieve spectral domain localization. The VGLCT is obtained as the IGLCT of $\hat{f}(z)$, localized using a spectral domain window $H(k - z)$ centered around the spectral index k [9]. Thus, the VGLCT can be expressed as follows

$$\mathcal{V}_h^A f(i, k) = \sum_{p=0}^{N-1} \sum_{l=0}^{N-1} \hat{f}_A(p) H(k - z) q_l(n) \kappa_l^* p_l^*(z) \epsilon_z^*. \quad (36)$$

It can be implemented using band-pass transfer functions, $H(k - z) = H_k(\lambda_z)$, as

$$\mathcal{V}_{h,2}^A f(i, k) = \sum_{p=0}^{N-1} \sum_{l=0}^{N-1} \hat{f}_A(z) H_k(\lambda_z) q_l(n) \kappa_l^* p_l^*(z) \epsilon_z^*. \quad (37)$$

The spectral domain localization window for a band-pass graph system can be expressed as a transfer function centered around eigenvalues λ_z and their surroundings [9], [12]. It is defined in polynomial form

$$H_k(\lambda_z) = h_{0,k} + h_{1,k}\lambda_z + \dots + h_{C-1,k}\lambda_z^{C-1} \quad (38)$$

where C represents the polynomial order, and $k = 0, 1, \dots, K-1$ with K being the number of bands. Then the elements $\mathcal{V}_{h,2}^A f(i, k)$, $i = 0, 1, \dots, N-1$ of the VGLCT matrix $\mathbf{V}_{h,2}^A$ can be written in a matrix form

$$\begin{aligned} \mathbf{v}_{h,2}^{k,A} &= \text{IGLCT}\{\hat{f}_A(z) H_k(\lambda_z)\} = \mathbf{Q}\mathbf{K}^{-1}\mathbf{P}^H\mathbf{Y}^{-1}\hat{\mathbf{f}}_A H_k(\Lambda_A) \\ &= \mathbf{Q}\mathbf{K}^{-1}\mathbf{P}^H\mathbf{Y}^{-1} H_k(\Lambda_A) \mathbf{Y}\mathbf{P}\mathbf{K}\mathbf{Q}^H \mathbf{f} \\ &= H_k(\mathcal{L}_A) \mathbf{f} = \sum_{m=0}^{C-1} h_{m,k} \mathcal{L}_A^m \mathbf{f} \end{aligned} \quad (39)$$

where $\mathbf{v}_{h,2}^{k,A}$, $k = 1, \dots, K-1$ is the column vector with elements $\mathcal{V}_{h,2}^A f(i, k)$, $i = 0, 1, \dots, N-1$, and $H_k(\Lambda_A)$ is a diagonal matrix with elements $H_k(\lambda_z)$, $z = 1, 2, \dots, N$. In this scenario, the number of shifted windows, K , is not related to the total number of indices N .

Based on this formulation of the VGLCT, we can further define two specific vertex-frequency analysis methods: the graph linear canonical wavelet transform (GLWT) and the graph linear canonical Stockwell transform (GLST).

1) *Graph Linear Canonical Wavelet Transform*: In classical signal processing, time-frequency analysis and wavelet transform share common goals but are typically treated independently. Time-frequency analysis explores the frequency and time characteristics of non-stationary signals, widely used in signal processing, communication, and audio analysis for tasks like parameter estimation, detection, and denoising. Wavelet analysis, offering a multi-resolution approach, captures both

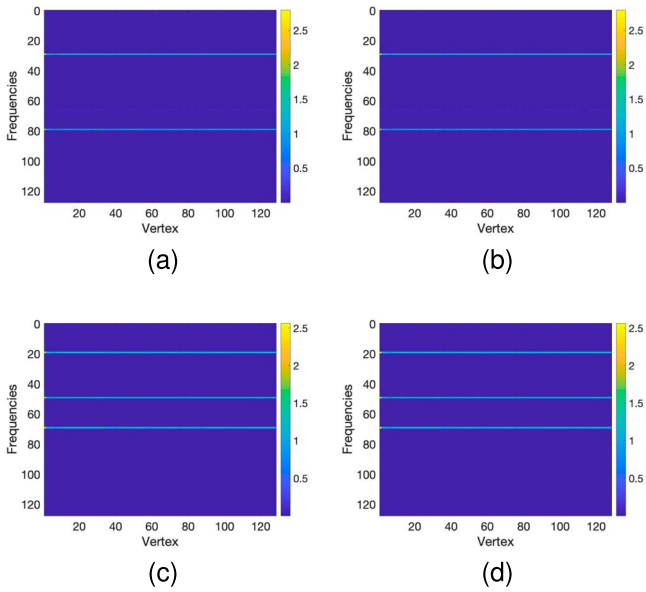


Fig. 10. (a) The WGLCT of $f_1(n)$ in Example 4 (duration: 21.043 seconds). (b) The fast WGLCT of $f_1(n)$ in Example 4 (duration: 0.265 seconds). (c) The WGLCT of $f_2(n)$ in Example 5 (duration: 20.691 seconds). (d) The fast WGLCT of $f_2(n)$ in Example 5 (duration: 0.195 seconds).

The detailed process is outlined in Algorithm 1.

Example 7: We utilize the signals $f_1(n)$ and $f_2(n)$ from Example 4 and Example 5, respectively ($\sigma = 1.5$, $\xi = 0$, and $\beta = 0.98$). As depicted in Fig. 10, it is evident that both the fast calculation and direct calculation methods yield the same results, indicating that they have an equivalent effect. The results

local and global signal features, enabling tasks such as signal compression, denoising, and feature extraction. In graph signal processing, it is often assumed that these two signal processing methods are equivalent. Therefore, only the GLWT, which is directly linked to frequency variation (VGLCT), is considered. It can be seen as a specific instance of vertex-frequency analysis, rather than a transform for graph signal compression and its wavelet-like multi-resolution analysis.

Graph linear canonical wavelet operators. Assuming a band-pass filter behavior, the wavelet definition in the spectral domain can be represented by $H(\lambda_z)$ as the basic form. We define the graph linear canonical wavelet operator at scale s_j as

$$\widehat{T_H^{s_j} f}(z) = H(s_j \lambda_z) \hat{f}(z) \quad (40)$$

where $j = 1, \dots, J$. This operator is achieved by applying each graph linear canonical mode to a given function f .

Employing the IGLCT on (40) yields

$$T_H^{s_j} f(z) = \sum_{z=0}^{N-1} \sum_{l=0}^{N-1} H(s_j \lambda_z) \hat{f}(z) q_l(n) \kappa_l^* p_l^*(z) \epsilon_z^*. \quad (41)$$

The graph linear canonical wavelets are then realized through localizing it by applying it to the impulse on a single vertex

$$\begin{aligned} \psi_{s_j, i}(n) &= T_H^{s_j} \delta_i(n) = \sum_{z=0}^{N-1} \sum_{l=0}^{N-1} H(s_j \lambda_z) \\ &\times \left(\sum_{r=0}^{N-1} \epsilon_k p_r(k) \kappa_r q_r^*(i) \right) q_l(n) \kappa_l^* p_l^*(z) \epsilon_z^*. \end{aligned} \quad (42)$$

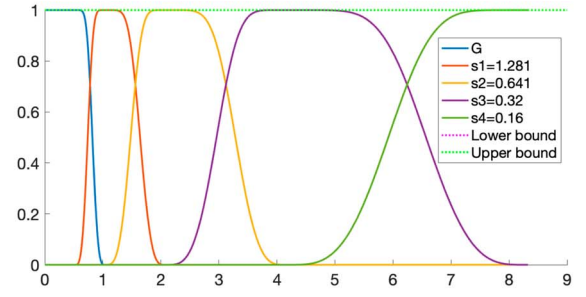
The GLWT is defined as

$$\begin{aligned} \mathcal{W}_\psi^A(i, s_j) &= \sum_{n=0}^{N-1} f(n) \psi_{s_j, i}(n) \\ &= \sum_{n=0}^{N-1} \sum_{p=0}^{N-1} \sum_{l=0}^{N-1} f(n) H(s_j \lambda_z) \\ &\times \left(\sum_{r=0}^{N-1} \epsilon_k p_r(k) \kappa_r q_r^*(i) \right) q_l(n) \kappa_l^* p_l^*(z) \epsilon_z^* \\ &= \sum_{p=0}^{N-1} \sum_{l=0}^{N-1} H(s_j \lambda_z) \hat{f}_A(z) q_l(n) \kappa_l^* p_l^*(z) \epsilon_z^*. \end{aligned} \quad (43)$$

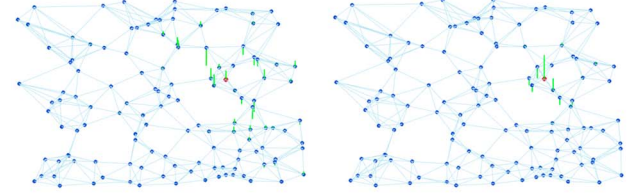
Example 8: We use wavelet generation kernels similar to Meyer wavelets [39], defined as

$$H(\lambda) = \begin{cases} \sin\left(\frac{\pi}{2}\nu\left(\frac{1}{\lambda_1}|\lambda| - 1\right)\right) & \text{if } \lambda_1 \leq \lambda \leq \lambda_2 \\ \cos\left(\frac{\pi}{2}\nu\left(\frac{1}{\lambda_2}|\lambda| - 1\right)\right) & \text{if } \lambda_2 \leq \lambda \leq \lambda_3 \end{cases} \quad (44)$$

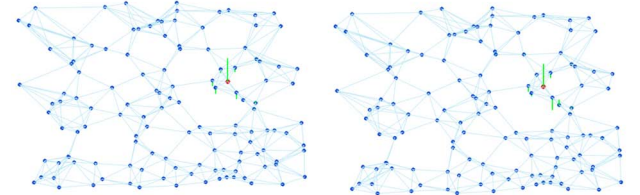
where $v(x) = x^4(35 - 84x + 70x^2 - 20x^3)$, and $\lambda_1 = 2/3$, $\lambda_2 = 2\lambda_1$, $\lambda_3 = 4\lambda_1$. The J wavelet scales are defined as $s_j = 2^j \lambda_{max}^{-1}$ for $j = 1, \dots, J$, where λ_{max} is obtained by the linear



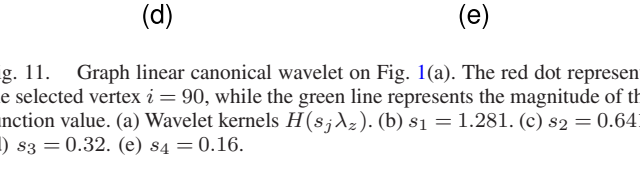
(a)



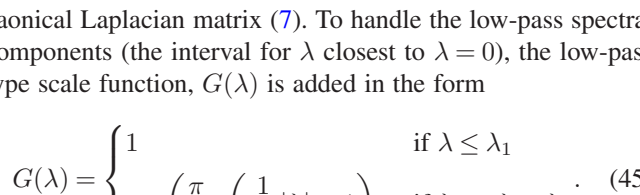
(b)



(c)



(d)



(e)

Fig. 11. Graph linear canonical wavelet on Fig. 1(a). The red dot represents the selected vertex $i = 90$, while the green line represents the magnitude of the function value. (a) Wavelet kernels $H(s_j \lambda_z)$. (b) $s_1 = 1.281$. (c) $s_2 = 0.641$. (d) $s_3 = 0.32$. (e) $s_4 = 0.16$.

canonical Laplacian matrix (7). To handle the low-pass spectral components (the interval for λ closest to $\lambda = 0$), the low-pass type scale function, $G(\lambda)$ is added in the form

$$G(\lambda) = \begin{cases} 1 & \text{if } \lambda \leq \lambda_1 \\ \cos\left(\frac{\pi}{2}\nu\left(\frac{1}{\lambda_1}|\lambda| - 1\right)\right) & \text{if } \lambda_1 \leq \lambda \leq \lambda_2 \end{cases}. \quad (45)$$

Let $J = 4$, the generated linear regular wavelet kernel is shown in Fig. 11. (a). Assuming values of $\sigma = 1$, $\xi = 0$ and $\beta = 0.98$, we can derive four scales: $s_1 = 1.281$, $s_2 = 0.641$, $s_3 = 0.32$, and $s_4 = 0.16$. By centering the wavelet on vertex 90, we can compute it and visualize it in Fig. 11(b)–11(e).

2) *Graph Linear Canonical Stockwell Transform:* The Stockwell transform is a time-frequency analysis technique that shares similarities with both the wavelet transform and the WFT. The Stockwell transform can be considered as an extension of the WT and WFT, offering a variable time-frequency resolution by using a variable-sized Gaussian window that adapts to the local characteristics of the signal.

Graph linear canonical modulation operator. For any signal $f(n)$ defined on the graph $\mathcal{G}$ and any $k = 0, 1, \dots, N - 1$,

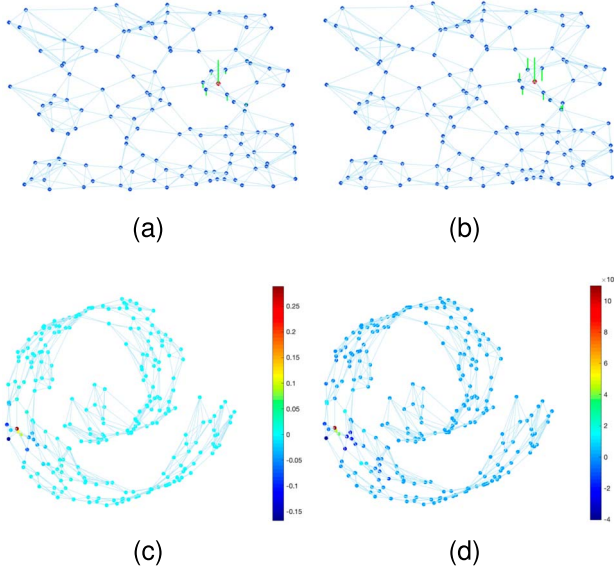


Fig. 12. (a) The GLWT kernel with $s_3 = 0.32, i = 90$ on Fig. 1. (a). (b) The GLST kernel with $s_3 = 0.32, i = 90$ on Fig. 1. (a). (c) The GLWT kernel with $s_3 = 0.396, i = 30$ on Fig. 1. (b). (d) The GLST kernel with $s_3 = 0.396, i = 30$ on Fig. 1. (b).

define the graph linear canonical modulation M_k^A by

$$M_k^A f(n) = \sum_{l=0}^{N-1} f(n) q_l(n) \kappa_l^* p_l^*(k) \epsilon_k^*. \quad (46)$$

The GLST can be defined as modulation of the GLWT [40]. For $s = 1$, denote its corresponding GLWT as $\mathcal{W}_\psi^A(i, s_j)$, the GLST is defined as

$$\begin{aligned} \mathcal{S}_h^A(i, s_j) &= \sum_{l=0}^{N-1} \mathcal{W}_h^A(i, s_j) q_l(i) \kappa_l^* p_l^*(k) \epsilon_k^* \\ &= \sum_{l=0}^{N-1} \sum_{z=0}^{N-1} \sum_{r=0}^{N-1} H(s_j \lambda_z) \hat{f}_A(p) q_r(n) \kappa_r^* p_r^*(z) \epsilon_z^* \\ &\quad \times q_l(i) \kappa_l^* p_l^*(k) \epsilon_k^*. \end{aligned} \quad (47)$$

Example 9: Briefly compare the differences in kernel functions between GLWT and GLST. Utilizing the basis function provided in Example 8, set the linear canonical parameter uniformly to $\sigma = 1, \xi = 0, \beta = 0.9$. Fig. 12 illustrates the outcome.

Example 10: Considering a path graph with trivial edge weights, we examine the signal $f_3 = \sin(60\pi n/N)$, where $0 \leq n \leq N-1, N = 256$. This signal represents a sampled sinusoidal waveform with a single frequency. Continuing with the utilization of the wavelet from Example 8, we set the parameters as $J = 4, \sigma = 2, \xi = 0$, and $\beta = 1$. As a result, we obtain four scales: $s_1 = 1.727, s_2 = 0.864, s_3 = 0.432$, and $s_4 = 0.216$. Fig. 13 depicts the GLWT and GLST of the signal $f_3(n)$ at scale $s_3 = 0.432$.

3) *Chebyshev Polynomial Approximation:* When employing a bandpass function in the windowing process and directly calculating it, conducting spectral analysis on the complete graph becomes imperative for deriving the basis function.

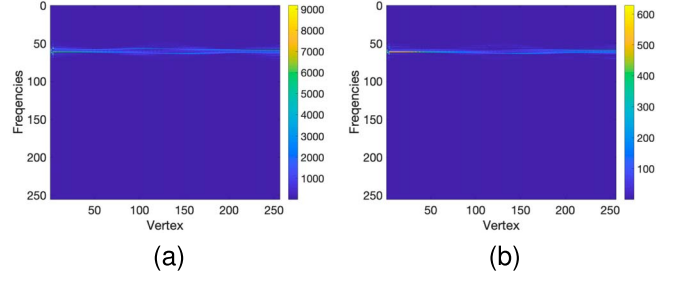


Fig. 13. (a) The GLWT of signal $f_3(n)$. (b) The GLST of signal $f_3(n)$.

To facilitate efficient computation, this section presents a polynomial approximation method grounded in the bandpass function $H_k(\lambda)$. This method enables vertex-frequency analysis using solely local signals and graph connections, thereby enhancing computational speed.

The discrete points $\lambda = \lambda_z$ are employed for the bandpass function $H_k(\lambda)$. However, since the polynomial approximation is continuous for $0 \leq \lambda \leq \lambda_{max}$, the “min-max” Chebyshev polynomial is naturally selected as it exhibits the least deviation from the desired bandpass function. This choice ensures the accuracy of the approximation. When the weight function is $\rho(x) = \frac{1}{\sqrt{1-x^2}}$, the orthogonal polynomial obtained by orthogonalizing the sequence $\{1, x, \dots, x^m, \dots\}$ is known as the Chebyshev polynomial, which can be represented as $T_m(x) = \cos(m \arccos x)$, where $|x| \leq 1$. The Chebyshev polynomial has the following important recursive relationships

$$\begin{aligned} T_{m+1}(x) &= 2xT_m(x) - T_{m-1}(x), \quad m = 1, 2, \dots \\ T_0(x) &= 1, \quad T_1(x) = x. \end{aligned} \quad (48)$$

By employing finite $(M-1)$ -order Chebyshev polynomials to approximate the spectral domain localization window in (38), i.e., $H_k(\mathcal{L}_A) = \tilde{P}_{k,M-1}(\mathcal{L}_A), k = 0, \dots, N-1$. The approximation has the form

$$\tilde{P}_{k,M-1}(\lambda) = \frac{c_{k,0}}{2} + \sum_{m=1}^{M-1} c_{k,m} \tilde{T}_m(\lambda) \quad (49)$$

where $\tilde{T}_m(\lambda) = \frac{2\lambda}{\lambda_{max}-1} T_m$ is utilized to map the argument from the interval $0 \leq \lambda \leq \lambda_{max}$ to the interval $[-1, 1]$. The Chebyshev coefficients are obtained by the Chebyshev polynomial inversion property, that is

$$c_{k,m} = \frac{\pi}{2} \int_{-1}^1 \frac{\tilde{P}_{k,M-1}(\lambda) \tilde{T}_m(\lambda)}{\sqrt{1-\lambda^2}} d\lambda. \quad (50)$$

This implementation is particularly noteworthy when the graph represents large-scale data.

Example 11: Considering the path graph and utilizing the signal $f_3(n)$ from Example 10, along with the wavelet mentioned in Example 8, the following parameters are used: $J = 6, \sigma = 1.5, \xi = 0$, and $\beta = 0.98$. Applying these parameters yields six scales: $s_1 = 6.744, s_2 = 3.372, s_3 = 1.686, s_4 = 0.843, s_5 = 0.422$, and $s_6 = 0.211$. Six filter kernels at different scales are obtained, as depicted in Fig. 14(a). Subsequently, these wavelet kernels are approximated using a 30th order Chebyshev polynomial, with the resulting approximation presented

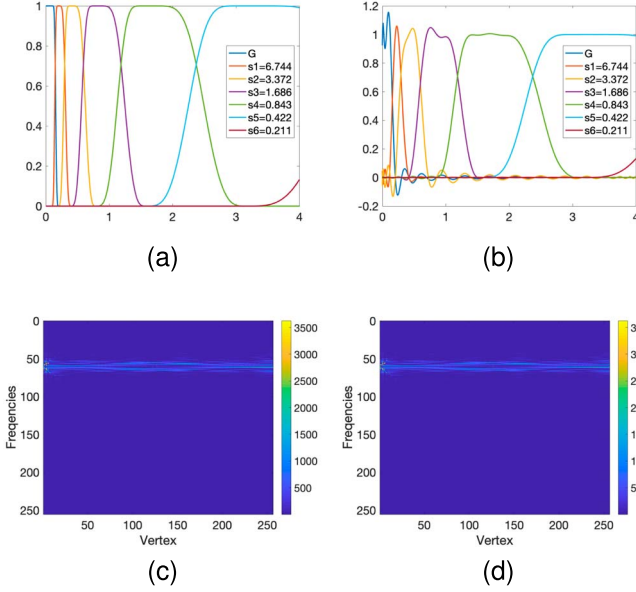


Fig. 14. (a) Wavelet kernels. (b) Approximated wavelet kernels. (c) The GLWT of $f_3(n)$ with (a) at scale $s_4 = 0.843$. (d) The GLWT of $f_3(n)$ with (b) at scale $s_4 = 0.843$.

in Fig. 14(b). Fig. 14(c) illustrates the vertex-frequency representation results using the original filter at scale $s_4 = 0.843$, while Fig. 14(d) displays the corresponding results using the approximation filter.

C. Localized Graph Linear Canonical Transform

The methods mentioned above all operate in the linear canonical domain of the graph. In this section, we discuss a vertex domain localization method denoted as localized graph linear canonical transform (LGLCT). Let $d_{n,i}$ represent the length of the shortest path from vertex i to vertex n . We then define the localized window function as $h(d_{n,i})$, where $h(\cdot)$ represents any basic window function commonly used in classical signal processing. For instance, the Blackman window can be employed [41], which is defined as follows

$$h(d_{n,i}) = 0.42 - 0.5\cos\left(\frac{2\pi d_{n,i}}{\eta}\right) + 0.08\cos\left(\frac{4\pi d_{n,i}}{\eta}\right) \quad (51)$$

where η is the assumed window width.

The distance $d_{n,i}$ can be calculated using the adjacency matrix $\mathbf{A}$. In an unweighted graph, a vertex belonging to a neighborhood of vertex i is represented by a unit value element in row i of the adjacency matrix $\mathbf{A}$. In the case of a weighted graph, the adjacency matrix $\mathbf{A}$ can be derived from the weighted matrix $\mathbf{W}$ by taking the sign of each element, i.e., $\mathbf{A} = \text{sign}(\mathbf{W})$. Then we define the following matrix form

$$\mathbf{A}_{d_{n,i}} = \begin{cases} \mathbf{A} & d_{n,i} = 1 \\ (\mathbf{A} \odot \mathbf{A}_{d_{n,i-1}}) \circ (\mathbf{1} - \mathbf{A}_{d_{n,i-1}}) \circ (\mathbf{1} - \mathbf{I}) & d_{n,i} \geq 2 \end{cases} \quad (52)$$

where $\odot$ represents the logical (Boolean) matrix product, $\circ$ denotes the Hadamard (element-by-element) product, and $\mathbf{1}$ is

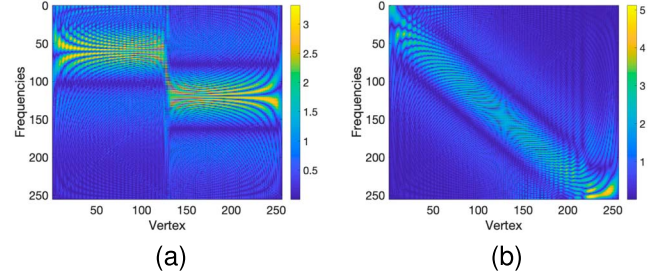


Fig. 15. The LGLCT of graph signals. (a) The LGLCT of signal $f_4(n)$. (b) The LGLCT of signal $f_5(n)$.

a matrix with all elements equal to 1. The i -th row of the matrix $(\mathbf{A} \odot \mathbf{A}_{d_{n,i}} - \mathbf{1})$ provides information about all vertices connected to vertex i through walks of length $K = d_{n,i}$ or lower. It is important to note that the element-wise multiplication of $(\mathbf{A} \odot \mathbf{A}_{d_{n,i}} - \mathbf{1})$ by the matrix $(\mathbf{1} - \mathbf{A}_{d_{n,i}} - \mathbf{1})$ removes the vertices connected by walks of length $d_{n,i} - 1$, while the multiplication by $(\mathbf{1} - \mathbf{I})$ eliminates the diagonal elements [9].

For a window function $\mathbf{h} = [h(0), \dots, h(N-1)]^T \in \mathbb{R}^N$, the localized window function (denoted as $\mathbf{T}_\mathbf{A}\mathbf{h} = [T_\mathbf{A}h(n, i)]_{N-1 \times N-1}$) with assumed graph window width η can be given by

$$\mathbf{T}_\mathbf{A}\mathbf{h} = h(0)\mathbf{I} + h(1)\mathbf{A}_1 + \dots + h(\eta-1)\mathbf{A}_{\eta-1}. \quad (53)$$

The LGLCT of a graph signal $f(n)$, then becomes

$$\mathcal{V}_{h,3}^A f(i, k) = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} f(n) T_\mathbf{A} h(n, i) \epsilon_k p_l(k) \kappa_l q_l^*(n). \quad (54)$$

Example 12: Given a path graph with trivial edge weights, we consider the following two time series: $f_4(n) = \begin{cases} \sin(60\pi n/N), & 1 \leq n \leq 128 \\ \sin(90\pi n/N), & 129 \leq n \leq 256 \end{cases}$ is a sampled sinusoidal signal with two different frequencies, $f_5(n) = \sin((20n + 0.5n^2)\pi/N), 1 \leq n \leq 256$ is a chirp signal. The LGLCT of signals $f_4(n)$ and $f_5(n)$ with parameters $\sigma = 1.05$, $\xi = 0$, and $\beta = 0.98$ is depicted in Fig. 15.

V. FILTER DESIGN

Within a basic vertex-frequency analysis framework, different filter selections can correspond to various transformations. The selection of an appropriate filter is crucial in practical applications. For instance, different spectral graph neural networks mainly rely on distinct filter learning approaches [26], [27], [28]. Therefore, in this section, optimization learning methods for filters are initially explored and subsequently applied to image classification tasks.

A. Optimal Filter

In Section IV-A, the graph linear canonical convolution operator was presented as $\mathbf{f} *_{\mathbf{A}} \mathbf{h} = \mathcal{F}_\mathbf{A}^H (\mathcal{F}_\mathbf{A} \mathbf{f} \odot \mathcal{F}_\mathbf{A} \mathbf{h})$, as shown in (25), where $\odot$ represents a point-wise product. In general, designing filters on a graph involves designing a diagonal matrix

$\mathbf{H}(\Lambda_A) = \text{diag}(\hat{\mathbf{h}})$, where $\hat{\mathbf{h}} = [h_1, h_2, \dots, h_N]^\top$. Then, the graph linear canonical convolution operator can be rewritten as

$$\mathbf{f} *_A \mathbf{h} = \mathcal{F}_A^H \mathbf{H}(\Lambda_A) \mathcal{F}_A \mathbf{f} = \mathbf{H}(\mathcal{L}_A) \mathbf{f}. \quad (55)$$

It follows that the signal $\mathbf{f}$ is filtered by $\mathbf{H}(\mathcal{L}_A)$.

Remark 2: Extending convolution to inputs $\mathbf{f}$ with multiple input channels is a straightforward process. If $\mathbf{f}$ is a signal with M input channels and N locations, we apply the GLCT $\mathcal{F}_A$ to each channel independently. Then, to perform the convolution, we utilize multipliers $\hat{\mathbf{h}} = (h_{i,j}; i \leq N, j \leq M)$.

Consider an input signal $\mathbf{f}$, where the target signal is represented as

$$\mathbf{g} = \mathbf{G}\mathbf{f} + \mathbf{n}, \quad (56)$$

where $\mathbf{G}$ is a known matrix, $\mathbf{n}$ is the additive noise term, and $\mathbf{f}$ is a stochastic graph signal [29]. The objective is to obtain an optimized filter that minimizes the loss function while predicting the output signal $\tilde{\mathbf{g}} = \mathbf{H}(\mathcal{L}_A)\mathbf{f}$. We define the squared loss function as $R(\mathbf{H}) = \frac{1}{N} \|\mathbf{H}(\mathcal{L}_A)\mathbf{f} - \mathbf{g}\|_F^2$. Therefore, the goal is to design the matrix $\mathbf{H}(\mathcal{L}_A)$ to solve the following optimization problem

$$\min_{\mathbf{H}} \frac{1}{N} \|\mathbf{H}(\mathcal{L}_A)\mathbf{f} - \mathbf{g}\|_F^2 \quad (57)$$

for any A .

Theorem 1: The problem defined in (57) will yield identical loss values for any A if there are no constraints on $\mathbf{H}(\mathcal{L}_A)$.

Proof: Consider any parameters A_1 and A_2 such that $A_1 \neq A_2$, and define

$$\mathbf{H}(\Lambda_{A_j}) = \arg \min_{\mathbf{H}} \left\{ \|\mathbf{H}(\mathcal{L}_{A_j})\mathbf{f} - \mathbf{g}\|_F^2 \right\} \quad (58)$$

where $j = 1, 2$. We note that for any choice of $\mathbf{H}(\Lambda_{A_j})$, according to $j = 1$ in (58), we have

$$\|\mathcal{F}_{A_1}^H \mathbf{H}(\Lambda_{A_1}) \mathcal{F}_{A_1} \mathbf{f} - \mathbf{g}\|_F^2 \leq \|\mathcal{F}_{A_1}^H \mathbf{H}(\Lambda_{A_2}) \mathcal{F}_{A_1} \mathbf{f} - \mathbf{g}\|_F^2.$$

Then, by choosing $\mathbf{H}(\Lambda) = \mathcal{F}_{A_1}^H \mathbf{H}(\Lambda_{A_2}) \mathcal{F}_{A_1}$, we can obtain from $j = 2$ in (58) that

$$\|\mathcal{F}_{A_1}^H \mathbf{H}(\Lambda_{A_1}) \mathcal{F}_{A_1} \mathbf{f} - \mathbf{g}\|_F^2 \leq \|\mathcal{F}_{A_2}^H \mathbf{H}(\Lambda_{A_2}) \mathcal{F}_{A_2} \mathbf{f} - \mathbf{g}\|_F^2.$$

Similarly, it can be easily shown that

$$\|\mathcal{F}_{A_2}^H \mathbf{H}(\Lambda_{A_2}) \mathcal{F}_{A_2} \mathbf{f} - \mathbf{g}\|_F^2 \leq \|\mathcal{F}_{A_1}^H \mathbf{H}(\Lambda_{A_1}) \mathcal{F}_{A_1} \mathbf{f} - \mathbf{g}\|_F^2.$$

Therefore, we can conclude that the same minimum value is achieved for any A_1 and A_2 . $\square$

Theorem 1 demonstrates that $\mathcal{F}_{A_1}$ and $\mathcal{F}_{A_2}$ can be interpreted as similarity transformation matrices. By deriving the objective function in (58), the optimal solution can be identified. Theorem 2 describes the solution to the optimal filtering problem posed in (58). Example 13 provides an illustration of an optimal filter.

Theorem 2: For the problem in (58), the optimal filter coefficients $\hat{\mathbf{h}}^{\text{opt}} = [h_1^{\text{opt}}, \dots, h_N^{\text{opt}}]^\top$ are obtained from the following matrix equation

$$(\mathcal{F}_A \mathbf{f} \odot \mathcal{F}_A \mathbf{f}) \mathbf{1}_N \odot \hat{\mathbf{h}}^{\text{opt}} = (\mathcal{F}_A \mathbf{f} \odot \mathcal{F}_A \mathbf{g}) \mathbf{1}_N. \quad (59)$$

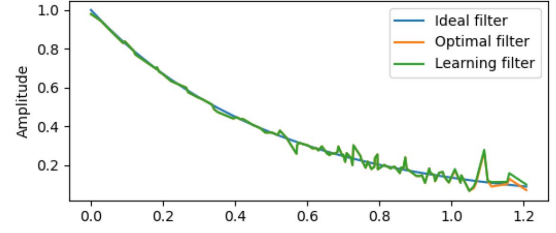


Fig. 16. Ideal filter, optimal filter, and learning filter.

Proof: Consider a fixed parameter A , the squared loss function is expressed as $R = \frac{1}{N} \|\mathcal{F}_A^H \mathbf{H}(\Lambda) \mathcal{F}_A \mathbf{f} - \mathbf{g}\|_F^2$. It is rewritten in the spectral domain as

$$\hat{R} = \frac{1}{N} \|\mathbf{H}(\Lambda) \mathcal{F}_A \mathbf{f} - \mathcal{F}_A \mathbf{g}\|_F^2. \quad (60)$$

Taking the derivative of $\hat{R}$ with respect to $\hat{\mathbf{h}}$, we have

$$\nabla_{\hat{\mathbf{h}}} \hat{R} = \frac{2}{N} \left(\mathcal{F}_A \mathbf{f} \odot (\hat{\mathbf{h}} \odot \mathcal{F}_A \mathbf{f} - \mathcal{F}_A \mathbf{g}) \right) \mathbf{1}_N. \quad (61)$$

Let $\nabla_{\hat{\mathbf{h}}} \hat{R} = 0$ and $\hat{\mathbf{h}}^{\text{opt}} = [h_1^{\text{opt}}, \dots, h_N^{\text{opt}}]^\top$, we obtain

$$(\mathcal{F}_A \mathbf{f} \odot \mathcal{F}_A \mathbf{f}) \mathbf{1}_N \odot \hat{\mathbf{h}}^{\text{opt}} = (\mathcal{F}_A \mathbf{f} \odot \mathcal{F}_A \mathbf{g}) \mathbf{1}_N. \quad (62)$$

$\square$

Example 13: Consider a K-NN graph with 100 vertices, where each vertex connects to 4 edges, where the spectral content of the graph signal length on each vertex follows a normal distribution, i.e., $\hat{f}_6(k) \sim \mathcal{N}(e^{-1.5 \cdot \lambda_k}, (e^{-1.5 \cdot \lambda_k} / 2)^2)$. Thus, the graph signal $\mathbf{f}_6 = \mathcal{F}_A^H \hat{\mathbf{f}}_6$ is a matrix of size 100×200 , where $f_6(n)$ denotes the signal on the n^{th} vertex, a vector with a length of 200. Let the parameters of GLCT be set to $\sigma = 1.5$, $\xi = 0.5$, $\beta = 0.95$. The ideal filter is defined as $h_k = e^{-2\lambda_k}$, where $k = 0, \dots, N-1$, and the noise term $\mathbf{n}$ follows a normal distribution $\mathbf{n} \sim \mathcal{N}(0, 0.1^2)$. Consequently, the actual output signal is given by $\mathbf{g} = \mathcal{F}_A^H \mathbf{H}(\Lambda) \mathcal{F}_A \mathbf{f}_6 + \mathbf{n}$. According to Theorem 2, the optimal filter design result can be derived. Fig. 16 illustrates the obtained optimal filter result depicted by the orange curve, while the blue curve represents the ideal filter.

B. Learning Filter

In the aforementioned optimal filter design method, the gradient of the loss function constitutes a non-convex function, rendering it challenging to guarantee the discovery of the global optimal solution. To accommodate diverse datasets and problem scenarios more flexibly, optimization algorithms like gradient descent can be employed for filter learning. This is particularly pertinent when dealing with more intricate models or elevated performance criteria. Furthermore, alongside the fundamental gradient descent method, numerous enhanced optimization algorithms exist, including stochastic gradient descent (SGD), the momentum method, Adam, Adagrad, and others. These algorithms offer various advantages tailored to different problem domains, aiding in expediting convergence and enhancing algorithmic performance.

Here, we introduce a method for learning filters via SGD, which serves as the groundwork for future advancements in

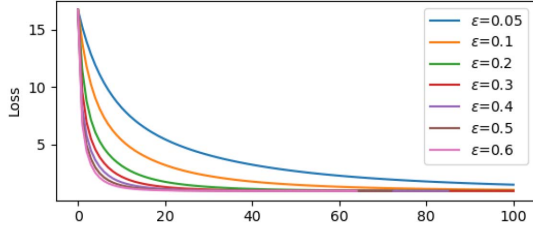


Fig. 17. Convergence analysis across different learning rates.

spectrogram neural networks, as shown in Algorithm 2. Assume the input signal $\mathbf{f}$ has dimensions $N \times M$, representing a graph with N vertices, where each vertex is associated with data of length M . The algorithm begins with the computation of $\mathcal{F}_A$, which has a computational complexity of $O(N^3)$. For each mini-batch, the complexity of gradient computation is $O(N_{\min}M)$, while the time complexity for updating the filter and computing the final filter's loss function is $O(N)$. Assuming the algorithm runs for T iterations and $N_{\min}$ remains constant, the primary computational cost is concentrated in the gradient computation and filter update during each iteration, with a complexity of $O(TN_{\min}M)$. Taking all factors into account, the overall time complexity of the algorithm is $O(TN_{\min}M + N^3)$.

As shown in Fig. 17, learning rates ϵ in the range of 0.3 to 0.6 lead to very rapid convergence, often within the first 20 iterations. While these higher learning rates provide faster convergence, the overlapping curves indicate that increasing the learning rate beyond $\epsilon = 0.3$ does not significantly improve the final outcome and may introduce risks of overshooting or instability in other scenarios. Based on this analysis, a learning rate in the range of 0.3 to 0.4 appears to be optimal for balancing fast convergence and stability. Therefore, a learning rate of 0.3 is selected, and Example 14 provides a specific instance of the learned filter. The results demonstrate the effectiveness of this approach.

Example 14: Continuing with the consideration of the graph signal $\mathbf{f}$ and target signal $\mathbf{g}$ in Example 13, we set the initial filter to a random sequence following a standard distribution, i.e., $\hat{\mathbf{h}}^0 \sim \mathcal{N}(0, 1)$. The parameters of GLCT are set to $\sigma = 1.5$, $\xi = 2$, and $\beta = 0.9$, with a learning rate of $\epsilon = 0.3$, a mini-batch size of $N_{\min} = 100$, a stopping criterion of 0.001, and a maximum number of iterations of 100. The final learned filter is depicted in Fig. 10, where the green curve represents the result.

C. Image Classification Based on Designed Filter

In addition to the direct utilization of advanced filters [32], [33], many researchers also employ polynomial form filters, which are extended in polynomial form (e.g., Chebyshev polynomial expansion as discussed in Section IV-B3). Polynomial filters offer the advantages of achieving locality and reducing learning complexity. Several notable examples include CheyNet [30], JacobiConv [28], LanczosNet [33], among others.

We focus on the benchmark MNIST classification problem [30], which consists of a dataset of 70,000 digits represented on a 28×28 grid. Particular emphasis is placed on distinguishing

Algorithm 2 SGD-based filter learning algorithm

Input:

Input graph signal: $\mathbf{f}$, Target signal: $\mathbf{g}$.
Initial value of filter: $\mathbf{H}^0$.
The parameters of GLCT: $A = (\xi, \sigma, \beta)$.
Learning rate: ϵ , Size of the mini-batch: $N_{\min}$, Stopping criterion, Maximum number of iterations.

Output:

The learned filter: $\mathbf{H}^{\text{sgd}} = \text{diag}(\hat{\mathbf{h}}^{\text{sgd}})$.
Loss function value: $\hat{R}(\mathbf{H}^{\text{sgd}})$.
1: Compute $\mathcal{F}_A = \Upsilon \mathbf{P} \mathbf{K} \mathbf{Q}^H$.
2: Compute the loss function in (60) of the initial filter: $\hat{R}(\mathbf{H}^0)$.
3: Let $\mathbf{H}^{\text{sgd}} = \mathbf{H}^0$.
4: **while** the stopping criteria are not met **do** :
Retrieve a mini-batch of samples with a length of $N_{\min}$ from the training set.
Calculate the gradient of the loss function: $\nabla_{\hat{\mathbf{h}}^{\text{sgd}}} \hat{R} = \frac{2}{N} (\mathcal{F}_A \mathbf{f} \odot (\hat{\mathbf{h}}^{\text{sgd}} \odot \mathcal{F}_A \mathbf{f} - \mathcal{F}_A \mathbf{g})) \mathbf{1}_N$.
Filter update: $\hat{\mathbf{h}}^{\text{sgd}} = \hat{\mathbf{h}}^{\text{sgd}} - \epsilon \nabla_{\hat{\mathbf{h}}^{\text{sgd}}} \hat{R}$.
end while
5: Compute the loss function in (60) of the learned filter: $\hat{R}(\mathbf{H}^{\text{sgd}})$.

TABLE I
PREDICTION RESULTS OF TEST SET WITH DIFFERENT PARAMETERS (%)

Method	Classifier 1 (GFT)	Classifier 2 (GFrFT)	Classifier 3 (GLCT)	Best Parameter (ξ, σ, α)
Chebyshev	85.39	85.77	89.39	(1, 1.25, 1.25)
Jacobi	89.38	89.33	90.00	(0, 1.5, 0.5)
Lanczos	90.28	92.44	94.44	(0.25, 1, 0.5)

between the digits 4 and 9, each comprising 6,824 samples, for a total of 13,648 instances, denoted as $M = 13,648$. These two digits are the most challenging to differentiate in the classification task. For the graph-based model, a 4-NN graph is constructed from the 2D grid, resulting in a graph with $N = |\mathcal{V}| = 784$ nodes. The weights of the K-NN similarity graph are computed using $w_{i,j} = \exp\left(-\frac{\|z_i - z_j\|_2^2}{\epsilon^2}\right)$, where z_i denotes the 2D coordinate of pixel i . Thus, the input signal is represented as $\mathbf{f} \in \mathbb{R}^{N \times M}$, while the target signal is represented as $\mathbf{g} \in \mathbb{R}^M$. Assuming the filter used for the classification task is denoted as $\mathbf{H}(\Lambda)$. To augment the fitting capability of the learner, a parameterized weight matrix Θ is introduced to perform radial transformations on the input graph signal matrix, yielding $\tilde{\mathbf{g}} = \mathcal{F}_A^H \mathbf{H}(\Lambda) \mathcal{F}_A \mathbf{f} \Theta$. The squared loss function in the spectral domain is given by

$$\hat{R} = \frac{1}{N} \|\mathbf{H}(\Lambda) \mathcal{F}_A \mathbf{f} \Theta - \mathcal{F}_A \mathbf{g}\|_F^2 \quad (63)$$

Subsequently, the filter undergoes training using Algorithm 2 to address image classification tasks.

For the dataset containing $M = 13,648$ samples, the data is randomly partitioned into three subsets: 70% for the training

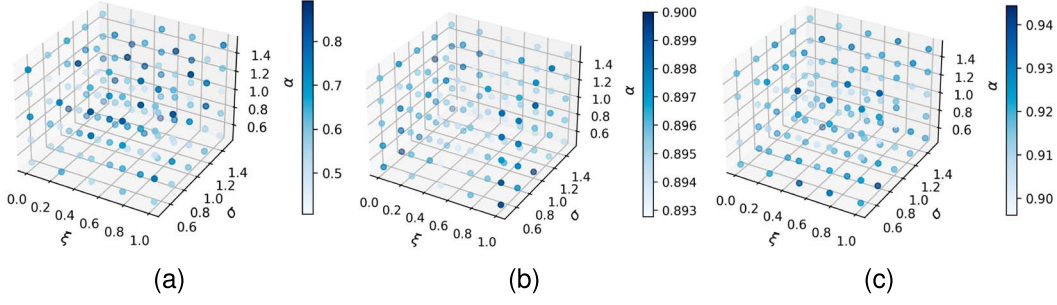


Fig. 18. The results under different parameters of Classifier 3. (a) Chebyshev. (b) Jacobi. (c) Lanczos.

set, 15% for the validation set, and 15% for the test set. The training set is used to fit the model, while the validation set is employed to fine-tune hyperparameters and adjust the model. Once the optimal parameters are determined, the test set is used to evaluate the final model performance and assess its generalization ability. To demonstrate the stability of the model, 10 independent training runs are performed, and the average result is used as the overall performance evaluation metric. We consider three different classifiers:

Classifier 1: The classifier uses GFT-based filters.

Classifier 2: The classifier uses GFrFT-based filters.

Classifier 3: The classifier uses GLCT-based filters.

Next, Chebyshev polynomials [30], Jacobi polynomials [28], and the Lanczos method [33] are used to evaluate the effectiveness of the proposed approach. As shown in Table I, $\alpha = 1.25$ is uniformly set as the parameter for Classifier 2, while the optimal parameter for GLCT is determined through a grid search, as illustrated in Fig. 18. Compared to Classifier 1, Classifier 3 achieves improvements of 4% and 4.16% using the Chebyshev polynomial and Lanczos method, respectively, and 0.62% using the Jacobi polynomial. By carefully selecting the parameters for GLCT, as illustrated in Fig. 18, classifier performance is further optimized. This underscores the flexibility and effectiveness of GLCT in improving model outcomes.

Remark 3: To enable comparison under a unified metric, this paper configures ChebyNet [30], JacobiNet [28], and LanczosNet [33] as single-layer filters. Additionally, a more intuitive evaluation is conducted to highlight the advantages of GLCT-based techniques over GFT-based approaches.

VI. CONCLUSION

This paper proposes a new graph linear canonical transform (GLCT). Furthermore, we discuss the vertex-frequency analysis framework based on this GLCT, delving into the specific delineations of different vertex-frequency analysis methods defined by various types of filters. Lastly, we examine filter design, encompassing optimal design and filter learning based on stochastic gradient descent, and apply them in image classification. The GLCT proposed in this paper demonstrates promising application prospects, with the filter design section akin to the convolutional layers of spectral graph neural networks (SGNNs), which can potentially be employed in large-scale SGNNs in the future.

APPENDIX GRAPH SCALING STAGE

The classical FT can be defined as

$$\mathcal{F}_c \Phi_k = e^{-ik\frac{\pi}{2}} \Phi_k \quad (64)$$

where Φ_k , $k = 0, 1, \dots$, are Hermite-Gaussian functions. The Hermite-Gaussian functions are defined as $\Phi_k(u) = \frac{1}{\sqrt{2^k k! \sqrt{\pi}}} e^{-\pi u^2} H_k(\sqrt{2\pi}u)$, where $H_k(u) = (-1)^k e^{u^2} \frac{d^k}{du^k} e^{-u^2}$ is the k^{th} -order Hermite polynomial. Starting with the Hermite-Gaussian differential equation [42]

$$\frac{d^2}{du^2} \Phi_k(u) - u^2 \Phi_k(u) = \lambda \Phi_k(u). \quad (65)$$

By replacing u with $\frac{u}{\sigma}$ in (65), i.e., adding scale parameters, and considering an operator $\mathcal{X}f(u) = uf(u)$, we obtain the following differential equation

$$(\sigma^4 \mathcal{F}_c \mathcal{X}^2 \mathcal{F}_c^{-1} + \mathcal{X}^2) \Phi_k \left(\frac{u}{\sigma} \right) = -\sigma^2 \lambda \Phi_k \left(\frac{u}{\sigma} \right) \quad (66)$$

where $\mathcal{F}_c$ represents the classical Fourier operator, utilizing the differential properties of the Fourier operator. Therefore, the discrete scale matrix is defined as $\mathbf{S}_{c\sigma} = \sigma^4 \mathcal{F}_c \mathbf{X}^2 \mathcal{F}_c^{-1} + \mathbf{X}^2$, where $\mathbf{X} = \text{diag}([0, 1^2, \dots, (N-1)^2])$ is an $N \times N$ matrix, and $\Phi_{k,\sigma}(u) = \Phi_k \left(\frac{u}{\sigma} \right)$ denotes the corresponding scaled eigenvectors. The scaled signal can be obtained by first decomposing the signal into the undilated Hermite eigenspace and then reconstructing it using dilated Hermite eigenvectors while preserving the original inner product coefficients. Specifically, this can be expressed as $f(n) = \sum_{k=0}^{N-1} c_k \Phi_k(n)$ and $f_\sigma(n) = \sum_{k=0}^{N-1} c_k \Phi_{k,\sigma}(n)$. Finally, the discrete scaling stage is defined as [36]

$$DS_\sigma = \mathbf{B}_\sigma \mathbf{B}^H \quad (67)$$

where $\mathbf{B}$ is an $N \times N$ matrix whose k^{th} column is the k^{th} order undilated Hermite eigenvector (i.e., $\Phi_k(u)$), and $\mathbf{B}_\sigma$ is an $N \times N$ matrix whose k^{th} column is the k^{th} order dilated Hermite eigenvector (i.e., $\Phi_k \left(\frac{u}{\sigma} \right)$).

For graph signals, the focus is on constructing a dilated graph eigenspace. By analyzing the properties of the dilated eigenspace in traditional signal processing, we expect the dilated eigenspace on the graph to satisfy the following properties:

- When $0 < \sigma < 1$, the dilated eigenspace is compressed, resulting in a narrower waveform and faster oscillation.

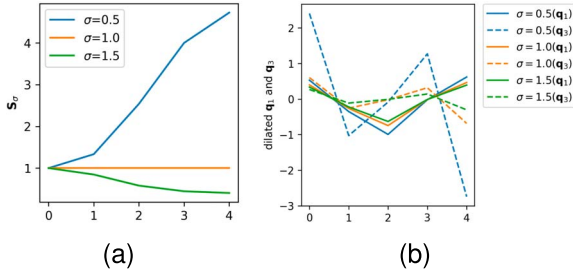


Fig. 19. (a) The scale matrix $\mathbf{S}_\sigma$. (b) The 1th and 3th dilated eigenvectors.

- When $\sigma = 1$, the dilated eigenspace is not scaled.
- When $\sigma > 1$, the dilated eigenspace expands, resulting in a wider waveform and slower oscillation.
- The larger the order k , the more pronounced the change, due to the properties of Hermite polynomials.

Therefore, we define a scale matrix as

$$\mathbf{S}_\sigma = \text{diag}([\sigma^{-\varepsilon r_0}, \sigma^{-\varepsilon r_1}, \dots, \sigma^{-\varepsilon r_{N-1}}]) \quad (68)$$

where r_l , $l = 0, \dots, N-1$, represents the eigenvalues corresponding to the undilated eigenspace $\mathbf{Q}$, and ε is a factor that represents the rate of scale change. The graph eigenspace dilated by this scale matrix satisfies the above four properties. The dilated graph eigenspace is obtained as $\mathbf{P} = \mathbf{S}_\sigma \mathbf{Q}$. Using Fig. 2(a) as an example, Fig. 19 shows the results of the scale matrix and the dilated eigenspace of the graph Fourier operator under different σ values. It can be observed that σ influences the compression and expansion effects.

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